

PROJECT REPORT

Real-Time Public Sentiment Dashboard (Twitter/X)

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Abstract

This project demonstrates the end-to-end development of a Natural Language Processing (NLP) pipeline designed to classify public sentiment from unstructured text. Utilizing the industry-standard Sentiment140 dataset comprising 1.6 million records, a custom machine learning model was trained to accurately categorize textual data into positive and negative sentiments. The final deliverable integrates Python-based predictive modeling with advanced data visualization via Power BI to provide actionable insights into public opinion, achieving a robust predictive accuracy of 79.4%.

Introduction

In the modern digital landscape, tracking public opinion regarding a brand, product, or service is critical for strategic decision-making. Millions of unstructured opinions are generated daily on social media platforms. The primary objective of this project is to capture raw, unstructured text and transform it into structured, quantifiable sentiment metrics.

While standard approaches often rely on basic word-matching dictionaries, this project implements a more sophisticated Machine Learning approach. By training a model to understand textual context and feature weightings, the resulting pipeline is far more resilient to the nuances, slang, and complexities of human language. The insights generated by the underlying model are subsequently visualized in an interactive dashboard to empower non-technical stakeholders.

Tools Used

- **Programming Language:** Python
- **Data Manipulation & Analysis:** Pandas, NumPy

- **Text Preprocessing:** Regular Expressions (re module)
- **Machine Learning Framework:** Scikit-Learn (Logistic Regression, TfidfVectorizer)
- **Development Environment:** Google Colab
- **Data Visualization & Dashboarding:** Power BI

Steps Involved in Building the Project

- **1. Data Acquisition & Handling:** The 1.6 million row dataset was programmatically sourced to bypass standard browser upload limitations and prevent data truncation, ensuring a perfectly balanced dataset of positive and negative records.
- **2. Text Preprocessing:** A custom text-cleaning function was engineered to sanitize the raw data. This involved converting text to lowercase and utilizing Regex to systematically strip out URLs, user mentions (@), and special characters.
- **3. Feature Extraction (TF-IDF):** The cleaned text was converted into a numerical format suitable for machine learning using a Term Frequency-Inverse Document Frequency (TF-IDF) Vectorizer. The vocabulary was capped at the top 10,000 most significant features to optimize computational performance.
- **4. Model Training & Validation:** The dataset was shuffled and split using an 80/20 training-to-testing ratio. A Logistic Regression classification model was trained on the high-dimensional data matrix. Logistic regression was selected for its exceptional speed and reliability in binary text classification tasks.
- **5. Model Evaluation & Custom Inference:** The model was evaluated against the testing set, achieving a 79.4% accuracy rate with evenly balanced precision and recall. A custom inference function was developed to allow real-time classification of new, user-generated text strings.
- **6. Dashboard Integration:** Model predictions on batch text data were structured for export to CSV, bridging the gap between the Python backend and Power BI. Visualizations were constructed to fulfill the final dashboarding requirement of the project phase.

Conclusion

This project successfully meets and exceeds the core objectives of tracking public sentiment. By independently building an NLP pipeline from scratch rather than relying on pre-packaged sentiment APIs, the project showcases an advanced understanding of data engineering, feature extraction, and predictive modeling. The seamless connection between the Python backend script and the

Power BI frontend visualization layer creates a highly accurate tool that translates unstructured public discourse into strategic business intelligence.